Lexture 09-The Repressilator

Repression must be strong enough
and the response slow enough
for an oscillator

$$\text{Re}\,\lambda_{2,3} = g/2 - \gamma > 0 \quad \Leftrightarrow \quad g/\gamma > 2$$

The local instability threshold of the symmetric three-variable model

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The quantitative argument

Strong enough

$g = -f'(x^*)$

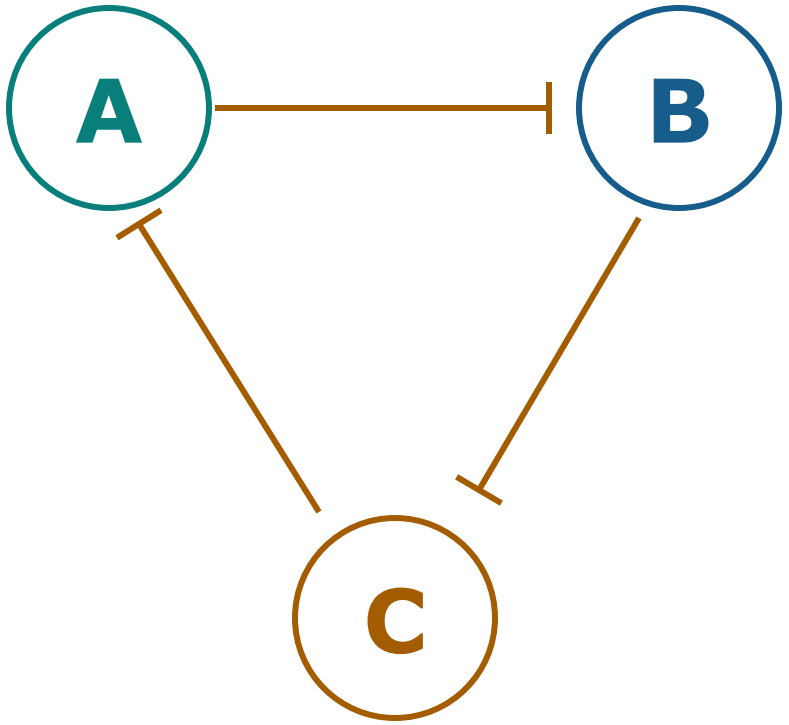
Production must respond steeply to upstream changes near x*.

Slow enough

$\tau = 1/\gamma$

Finite turnover provides memory. Lower γ weakens damping at fixed g.

Three repressors in a negative-feedback ring



A high represses B

Low B allows C to rise

High C represses A

Expression and turnover take time.
Previous molecules persist as
the next phase develops.

The reduced biological model

$$\frac{dx_i}{dt} = \alpha / (1 + x_{i-1}^n) + \alpha_0 - \gamma x_i$$

x_i is a repressor concentration scaled by a common repression threshold.

Parameter	Biological meaning
α, n	Regulated production range and Hill steepness
α_0	Basal leak expression under repression
γ	Degradation plus growth dilution

The reduced model has one variable per node. The original circuit model separates mRNA and protein.

Two-stage and three-stage repression rings

Two stages

A ⊣ B ⊣ A

(−) × (−) = (+)

Positive feedback can support mutually exclusive states or bistability.

Three stages

A ⊣ B ⊣ C ⊣ A

(−) × (−) × (−) = (−)

Negative feedback plus response lag can support oscillatory instability.

Topology supplies the feedback sign. Kinetics and eigenvalues determine the instability.

Why the simple two-variable model excludes Hopf

$dx/dt = F(y) - \gamma_x x$

$dy/dt = G(x) - \gamma_y y$

$$J = \begin{pmatrix} -\gamma_x & F'(y^*) \\ G'(x^*) & -\gamma_y \end{pmatrix}$$

$\text{tr } J = -(\gamma_x + \gamma_y) < 0$

$\det J = \gamma_x\gamma_y - F'(y^*)G'(x^*)$

In two dimensions, Hopf requires a zero trace and positive determinant.

Negative loop: $F'G' < 0$ gives $\det J > 0$. Double repression can instead lose stability at $\det J = 0$.

The production function and the equilibrium

Production function

$$f(x) = \alpha / (1 + x^n) + \alpha_0$$

Complete dynamics, with $x_0 = x_3$

$$dx_i/dt = f(x_{i-1}) - \gamma x_i$$

$$x_1 = x_2 = x_3 = x^*$$

$$f(x^*) = \gamma x^*$$

Identical parameters and the symmetric steady state make the common slope g well defined.

Linearization exposes gain and removal

$x_i = x^* + u_i$

$f(x^* + u_{i-1}) \approx f(x^*) + f'(x^*)u_{i-1}$

$du_i/dt \approx [f(x^*) - \gamma x^*] + f'(x^*)u_{i-1} - \gamma u_i$

The equilibrium terms cancel.

$$du_i/dt = -\gamma u_i - g u_{i-1}$$

$g = -f'(x^*) = \alpha n (x^*)^{n-1} / [1 + (x^*)^n]^2 > 0$

The Jacobian records the cyclic inhibition

$$du/dt = Ju$$

$$J = \begin{bmatrix} -\gamma & 0 & -g \\ -g & -\gamma & 0 \\ 0 & -g & -\gamma \end{bmatrix}$$

Diagonal entries

$-\gamma$ removes each node's deviation.

Cyclic off-diagonal entries

$-g$ transmits the upstream inhibitory response.

$u = (u_1, u_2, u_3)^T$. The missing cross-links have zero derivatives.

The characteristic equation

$u(t) = v \exp(\lambda t)$ gives $Jv = \lambda v$

$\lambda I - J =$

$$\begin{bmatrix} \lambda+\gamma & 0 & g \\ g & \lambda+\gamma & 0 \\ 0 & g & \lambda+\gamma \end{bmatrix}$$

$$\det(\lambda I - J) = (\lambda + \gamma)^3 + g^3 = 0$$

A nonzero v requires a zero determinant. The two surviving products come from the diagonal and the ring.

The complex cube roots supply the factor 2

$z = (\lambda + \gamma)/g$ gives $z^3 = -1$

$z_1 = -1, \quad z_{2,3} = 1/2 \pm i\sqrt{3}/2$

$\lambda_1 = -\gamma - g$

$\lambda_{2,3} = -\gamma + g/2 \pm i\sqrt{3}g/2$

The factor 2 comes from the real part 1/2 of the complex cube roots. Feedback must supply $g/2 > \gamma$.

Real part controls growth, imaginary part controls frequency

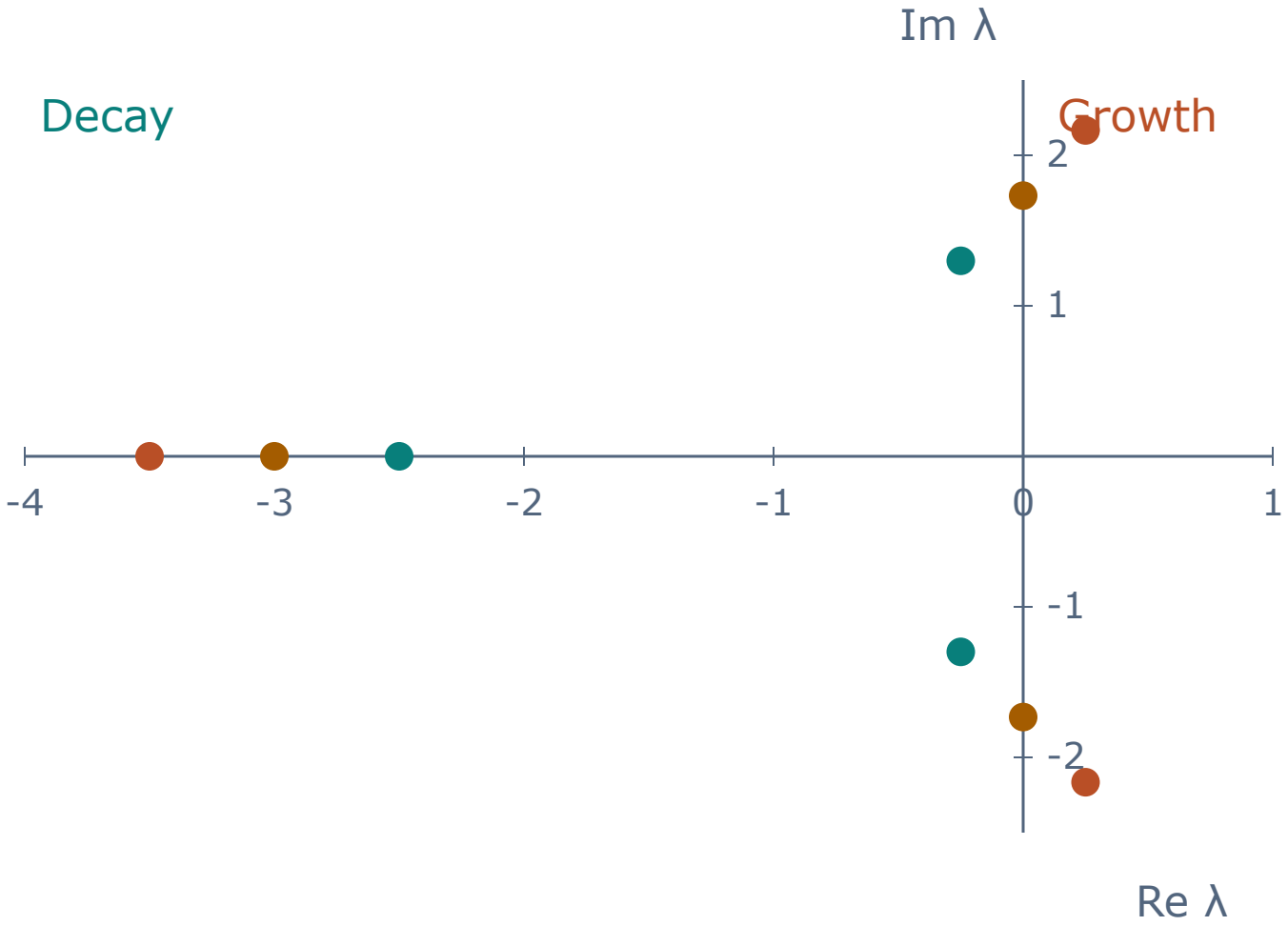
$u(t) = A \exp(at) \cos(\omega t + \phi)$

$a = g/2 - \gamma$ $\omega = \sqrt{3}g/2$

Condition	a	Local response
$g/\gamma < 2$	< 0	All perturbations decay
$g/\gamma = 2$	$= 0$	Candidate Hopf boundary
$g/\gamma > 2$	> 0	Oscillatory perturbations grow

At onset: $\omega_H = \sqrt{3}\gamma$, $T_H = 2\pi/(\sqrt{3}\gamma)$

The oscillatory pair crosses the imaginary axis



γ = 1.0 in each example

g = 1.5

Re $\lambda_{2,3} = -0.25$

g = 2.0

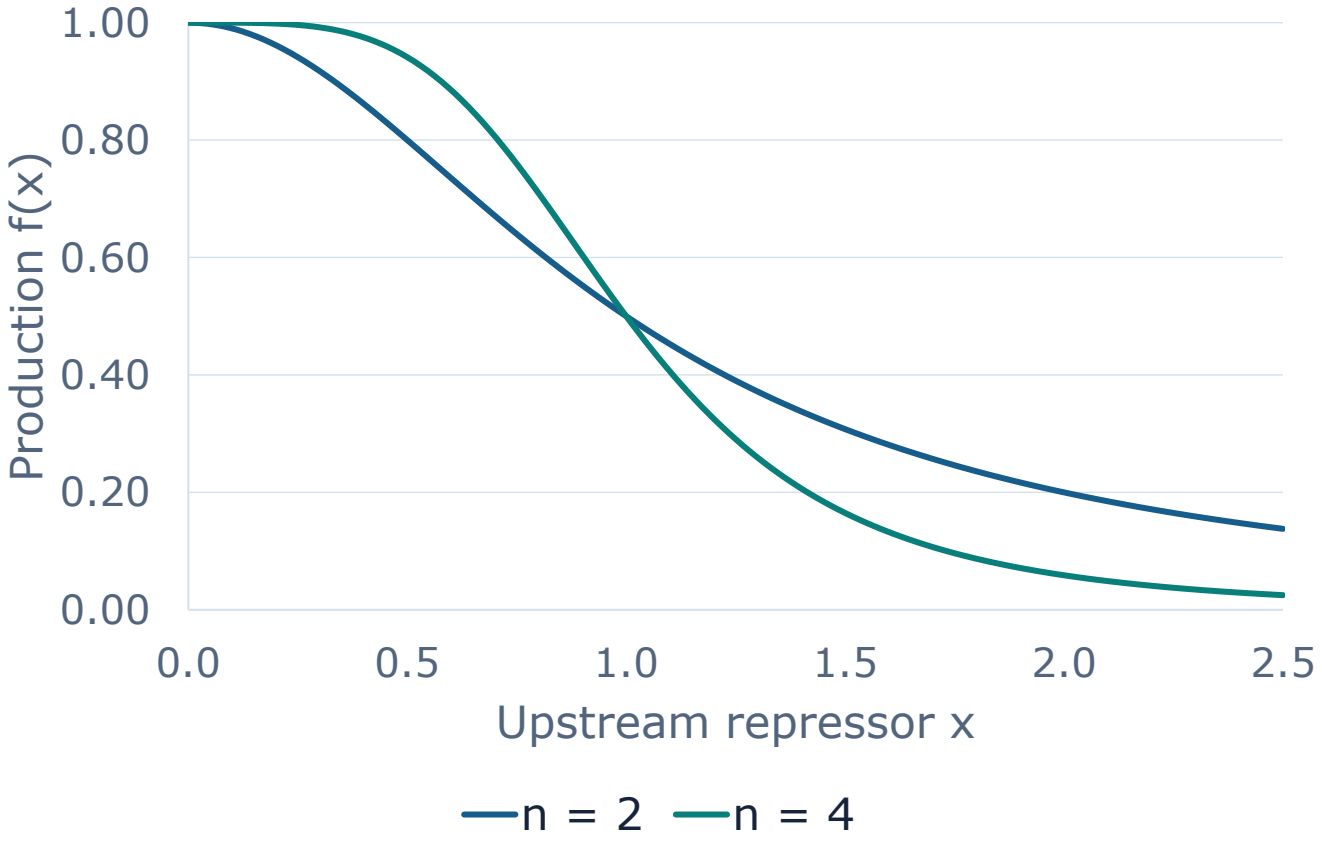
Re $\lambda_{2,3} = 0$

g = 2.5

Re $\lambda_{2,3} = +0.25$

At fixed γ, increasing g moves the pair right. At fixed g, decreasing γ also moves it right.

Large g means a steep local repression slope



$g = -f'(x^*)$

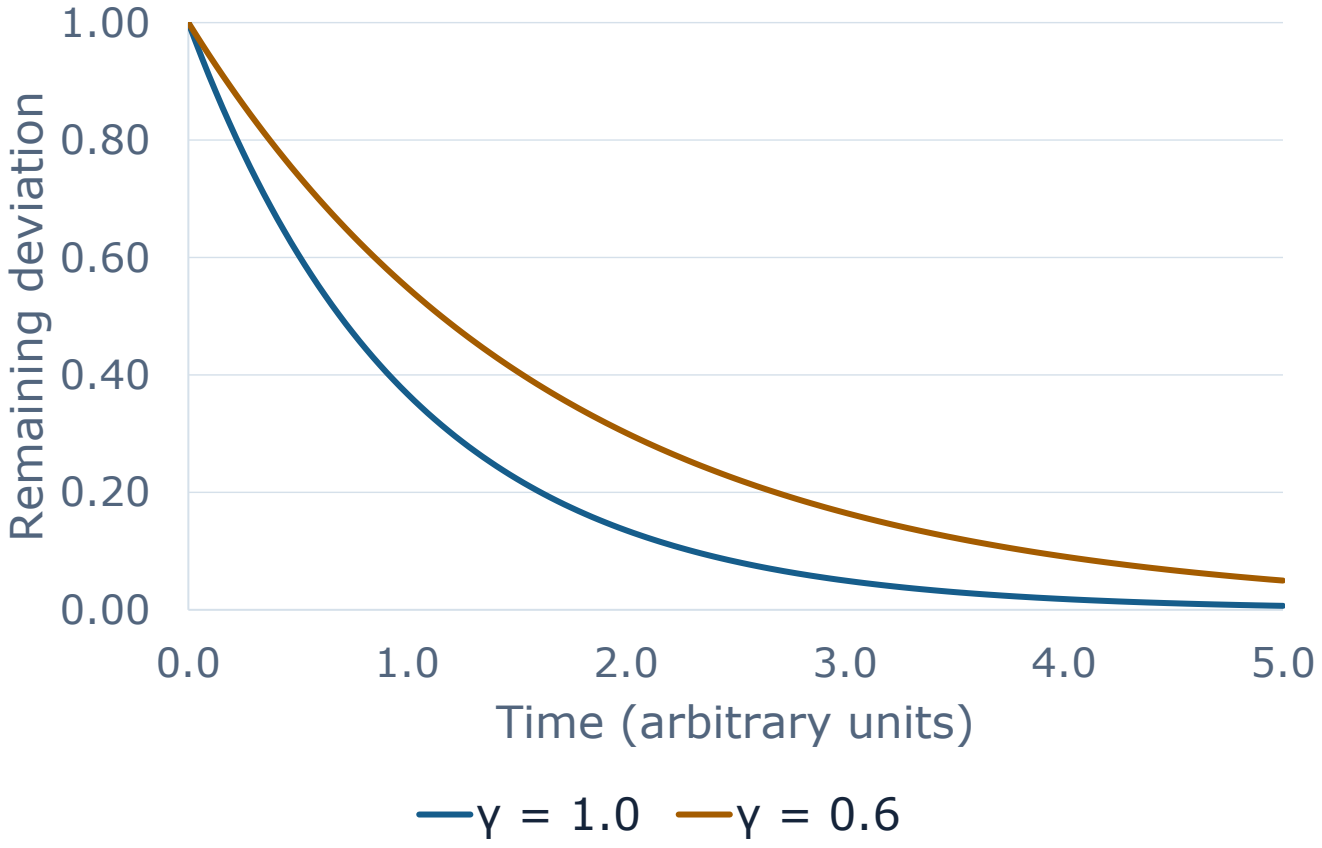
Local regulatory sensitivity

Cooperativity, production range, leak and x* affect g.

Measure promoter production and its slope near x*.

Illustrative curves: $f(x) = 1/(1 + x^n)$. At $x^* = 1$, the n = 4 curve has twice the slope magnitude of n = 2.

Small γ means longer concentration memory



$\gamma = \gamma_{deg} + \gamma_{dil}$

Degradation plus growth dilution

$\tau = 1/\gamma$

Half-life = $\ln 2/\gamma$

Longer persistence adds memory.

Measure protein clearance and cell growth dilution.

Illustrative removal alone: $u(t)/u(0) = \exp(-\gamma t)$. Lower γ weakens damping when g is held fixed.

Local examples compare gain with damping

Local example	g	γ	g/γ	Re $\lambda_{2,3}$	Outcome
Reference	1.5	1.0	1.5	−0.25	Stable
Steeper repression	2.5	1.0	2.5	+0.25	Unstable
Slower removal	1.5	0.6	2.5	+0.15	Unstable
All rates doubled	3.0	2.0	1.5	−0.50	Still stable

A larger ratio can destabilize the steady state.
Doubling both rates preserves the stability sign.

Arbitrary inverse-time units. These are local linear coefficients, not fitted biological measurements.

Strong enough and slow enough

The ring supplies negative feedback.

The local slope supplies gain g.

Turnover supplies damping γ and memory 1/γ.

$$\mathrm{Re}\,\lambda_{2,3} = g/2 - \gamma > 0 \quad \Leftrightarrow \quad g > 2\gamma$$

The nonlinear system determines the stable cycle.
Period sensitivity and timing noise determine its reliability.